One of the neatest examples of this concerns the phenomenon known as *mimicry*. Some butterflies taste nasty. They are usually brightly and distinctively coloured, and birds learn to avoid them by their 'warning' marks. Now other species of butterfly that do not taste nasty cash in. They *mimic* the nasty ones. They are born looking like them in colour and shape (but not taste). They frequently fool human naturalists, and they also fool birds. A bird who has once tasted a genuinely nasty butterfly tends to avoid all butterflies that look the same. This includes the mimics, and so genes for mimicry are favoured by natural selection. That is how mimicry evolves.

There are many different species of 'nasty' butterfly and they do not all look alike. A mimic cannot resemble all of them: it has to commit itself to one particular nasty species. In general, any particular species of mimic is a specialist at mimicking one particular nasty species. But there are species of mimic that do something very strange. Some individuals of the species mimic one nasty species; other individuals mimic another. Any individual who was intermediate or who tried to mimic both would soon be eaten; but such intermediates are not born. Just as an individual is either definitely male or definitely female, so an individual butterfly mimics either one nasty species or the other. One butterfly may mimic species *A* while his brother mimics species *B*.

It looks as though a single gene determines whether an individual will mimic species *A* or species *B*. But how can a single gene determine all the multifarious aspects of mimicry—colour, shape, spot pattern, rhythm of flight? The answer is that one gene in the sense of a *cistron* probably cannot. But by the unconscious and automatic 'editing' achieved by inversions and other accidental rearrangements of genetic material, a large cluster of formerly separate genes has come together in a tight linkage group on a chromosome. The whole cluster behaves like a single gene—indeed, by our definition it now *is* a single gene—and it has an 'allele' which is really another cluster. One cluster contains the cistrons concerned with mimicking species *A*; the other those concerned with mimicking species *B*. Each cluster is so rarely split up by crossing-over that an intermediate butterfly is never seen in nature, but they do very occasionally turn up if large numbers of butterflies are bred in the laboratory.

I am using the word gene to mean a genetic unit that is small enough to last for a large number of generations and to be distributed around in